Bukang-liwayway v3.1g — Printing & Assembly Guide

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Before you start

Bukang-liwayway ("Sunrise") is Sol Prosthetics' Model 1: a wrist-driven hand for children about 9 and under. Bending the wrist down pulls four braided lines and the fingers close; bending it back lets elastic cords open them. The thumb sits on a friction hinge you set by hand.

- **This is a test print.** v3.1g has not yet been printed or fitted to a child. Treat the first one as a trial and check everything in the Tuning section.
- **Print at 100% only.** The screw holes are sized for M3 and come out wrong if you scale the files in the slicer.
- **It fits** a child's hand about 55 mm across the knuckles and 129 mm from wrist crease to middle fingertip, with a palm up to about 49 mm long from the wrist crease. Wrist about 97 mm round, forearm about 110 mm.
- **Materials:** PETG for every rigid part (never PLA, it softens in the heat) and TPU 95A for the palm cover.
- If you can, fit the first hands with a physical therapist or prosthetist. Every residual limb is different.

Parts to print

There are 15 files in the zip's `stl` set, one of each part. Every file is already turned to its best print direction, so load it as it is; don't rotate it.

File	Part	Material	Size (mm)	Supports
`palm_back.stl`	Back of the hand: knuckles, thumb mount, wrist ears, line tunnels, logo, step for the cover	PETG	73 × 51 × 78	Yes, from the build plate. It stands on the wrist end; use a brim

File	Part	Material	Size (mm)	Supports
`cuff.stl`	Forearm cuff with tensioner swelling, line clamps, strap slots, vents	PETG	61 × 39 × 110	A few
`palm_flap.stl`	Palm cover with raised grip oval, edge lip, 7 snap pegs, 3 screw holes	**TPU 95A**	39 × 17 × 49	A few, under the snap pegs
`thumb.stl`	Thumb with hinge tongue, fingertip recess and nail	PETG	17 × 39 × 25	A few
`finger-index-1/2/3.stl`	Index finger, 3 segments	PETG	up to 19 × 37 × 16	A few
`finger-middle-1/2/3.stl`	Middle finger, 3 segments	PETG	up to 23 × 41 × 16	A few
`finger-ring-1/2/3.stl`	Ring finger, 3 segments	PETG	up to 21 × 39 × 15	A few
`finger-little-1/2.stl`	Little finger, 2 segments	PETG	up to 18 × 30 × 14	A few

Segment 1 is the one that joins the palm; segment 3 (2 on the little finger) is the fingertip with the nail. Plastic needed: roughly 100–135 g of PETG plus supports, and under 10 g of TPU.

Print settings

A 0.4 mm nozzle and 0.2 mm layers are assumed for every part.

Setting	PETG parts	TPU palm cover
Walls	5	4, so the 3 mm snap pegs print solid
Top / bottom layers	Your usual	4 each

Setting	PETG parts	TPU palm cover
Infill	40–50% gyroid (fingers, thumb, back of hand); 30% (cuff)	30% gyroid (the 2 mm cover is almost all walls and top/bottom layers)
Speed	Normal for your PETG	Slow, 20–30 mm/s
Retraction	Normal	Little or none
Surface	Normal	**Fuzzy skin** on the outer grip face only, so the palm isn't slippery. Never on the snap pegs: fuzzy skin adds about 0.3 mm and the 3.0 mm pegs then jam or tear in their 3.2 mm holes. Paint it on the grip face, or use a modifier to turn it off round the pegs. If that's a bother, skip it; TPU 95A is grippy anyway.
Supports	Touching the build plate only (back of the hand); a few on the rest	Under the snap pegs
Brim	Back of the hand	Optional

Per-part tips:

- **Back of the hand:** it stands on its wrist end. Supports never touch the logo, so the logo comes out clean.
- **Fingers and thumb:** they lie with the screw holes vertical, so every hole prints round and the layers run along the finger for strength. Don't turn them.
- **All parts:** every file is already a clean, closed solid. Don't run the slicer's mesh-repair, "fill holes" or "remove small features" options on it.
- **Palm cover:** dry the TPU first if it has been sitting out; wet TPU strings and blobs.

Hardware and materials

All screws are stainless M3 from a hardware store. No nuts are needed: every screw threads into a 2.6 mm pilot hole in the plastic.

Item	Qty	Where it goes
M3 × 12 **button-head** screw (ISO 7380)	9	Index and middle: knuckle, middle and tip joints; ring: middle and tip joints; thumb hinge

Item	Qty	Where it goes
M3 × 10 button-head screw	3	Ring knuckle; little finger knuckle and tip joint
M3 × 6 screw	2	Wrist hinge, one through each cuff arm
M3 × 4 **countersunk (flat-head)** screw	4	Line clamps on top of the cuff
M3 × 5 **countersunk (flat-head)** screw	3	The palm cover's little-finger edge
Braided fishing line, about 50 lb / 20 kg	about 2 m	4 flexor lines (close the fingers)
1 mm round elastic cord	about 1.5 m	4 return cords (open the fingers)
Bicycle inner tube	1 small piece	5 grip pads and 2 thumb-hinge washers
Contact cement (e.g. Rugby)	a little	Gluing the grip pads
Sew-on hook-and-loop strap, 25 mm	2	Cuff straps
Washable foam or mesh liner, 1–1.5 mm	1 piece	Inside the cuff and the back of the hand

Tools you need

- 2 mm hex key (for M3 button-head and flat-head screws), with a long one or a hex bit for the knuckles
- Flush cutters and a craft knife for removing supports and brims
- Fine sandpaper or a needle file
- Thin wire or a long darning needle for threading lines through the tunnels
- Lighter or a drop of super glue for sealing knots in the braided line
- Scissors, rubbing alcohol and a cloth
- Needle and strong thread for sewing on the straps

After printing

- [ ] Remove all supports and the brim. Pay attention inside the knuckle cups, the thumb mount slot and under the cover's snap pegs.

- [ ] Sand smooth any edge that will touch skin: the wrist end of the back, the cuff ends and strap slots.
- [ ] Run an M3 screw through every joint hole once to cut the threads, then back it out.
- [ ] Push a piece of line through every channel and tunnel to check it is clear: the upper and lower channel in every finger segment, 8 tunnels in the back of the hand, 4 in the cuff.
- [ ] Check that each of the 7 snap holes in the back's step is clear, with its small round recess on the inside.
- [ ] Dry-fit the cover in its step: it should lie flush all round with no gaps.

Assembly

Work in this order; each step leaves the next screw reachable. Tighten joints only until they move freely without wobble.

1. **Build each finger.** Join its segments in order (1 to 2, 2 to 3). Each joint is a rounded barrel in a cup held by one button-head screw. Put the screw in from the side with the round recess, so the head sits flush; it threads into the pilot hole on the far side. A flat stop on the back of each joint keeps the finger from bending backwards.
   - Index: M3 × 12 (middle joint), M3 × 12 (tip joint)
   - Middle: M3 × 12 (middle joint), M3 × 12 (tip joint)
   - Ring: M3 × 12 (middle joint), M3 × 12 (tip joint)
   - Little: M3 × 10 (its one joint)
2. **Fit the fingers to the palm: middle, then index, ring, little.** Slide segment 1's cup over its knuckle tongue on the back of the hand and screw it in from the recessed side: M3 × 12 for index and middle, M3 × 10 for ring and little. The middle finger goes first so every screw is still reachable.
3. **Thumb.** Cut two small washers from inner tube. Put one on each side of the thumb's tongue, fit it into the slotted barrel on the thumb mount and screw in the M3 × 12. Swing the thumb to the grip position you want, then tighten until it holds there.
4. **Return cords (open the fingers).** Knot the elastic in the knot pocket under the fingertip recess. Thread it through the upper channel of each segment and through its tunnel in the back of the hand. Knot it in the small counterbore at the wrist end of the back, just snug enough to hold the finger open.
5. **Flexor lines (close the fingers).** Knot each braided line in the same fingertip pocket and seal the knot. Thread it through the lower channel of each segment, into the knuckle block and through its tunnel to the wrist end. A thin wire or long needle helps.
6. **Wrist.** Join the back of the hand to the cuff with the two M3 × 6 screws through the cuff's hinge arms into the hand's ear pads.

7. **Tensioner.** Feed each flexor line into its hole in the front end of the cuff. It runs inside the wall, under its clamp, and comes out on the inside of the cuff behind the swelling. With the wrist straight and the fingers open, pull it snug and screw in its M3 × 4 flat-head clamp screw from the top until it sits flush; the screw pins the line. Trim the tail inside the cuff.
8. **Grip pads.** Cut inner tube to the recess sizes below. Wipe the recess and the rubber with alcohol, put a thin film of contact cement on both, let it go tacky (about 10 minutes), then press the pad in firmly.
9. **Palm cover.** Lay the cover's little-finger edge in its step and fit the three M3 × 5 flat-head screws from outside until they sit flush. Don't overtighten; the TPU only needs to be held. Fold the cover shut and press firmly over each of the 7 snap pegs until its head pops through the hole in the step. You'll feel each one click.
10. **Straps and liner.** Sew the two straps onto the cuff through its strap slots, fit the liner inside the cuff and the back of the hand, and sand any edge that touches skin.

Grip pad	Cut size (mm)
Index fingertip	7 × 6.5
Middle fingertip	7 × 7
Ring fingertip	7 × 6.5
Little fingertip	7 × 9
Thumb tip	11 × 8

Tuning and test checks

In CAD, bending the wrist down 40° closes the fingers about 43–48° per joint, and the thumb meets the index fingertip at about 48° down. Check the real print against that.

- [ ] **Line tension:** bend the wrist down and watch all four fingers. If one closes late or not at all, loosen its clamp screw, pull its line a little tighter and re-tighten. Adjust until all four close together.
- [ ] **Opening:** with the wrist straight, every finger should spring fully open. If one stays bent, tighten its return cord knot at the wrist end.
- [ ] **Joint friction:** each joint should swing freely with no side wobble. Loosen or tighten its screw by a quarter turn at a time.
- [ ] **Pinch:** close the hand and check that the thumb meets the index fingertip. For an earlier, tighter pinch, swing the thumb a little further on its hinge and re-tighten.

- [ ] **Wrist range:** the hand should swing from about 55° down to 25° up without rubbing the cuff.
- [ ] **Cover:** open and close it ten times. Every snap should click in and the edge should stay flush.
- [ ] **Fit:** with the liner in, the child's hand should slide in from the palm side without forcing, and nothing should press on the skin.

Putting it on and daily care

1. Peel the cover open from the thumb side; it stays screwed on along the little-finger edge.
2. Slide the child's hand in from the palm side.
3. Press the cover shut over each snap until it clicks.
4. Do up the two cuff straps, snug but not tight.

- **Check the child's skin every day,** especially in the heat. A closed shell hides red marks and pressure spots. If a mark doesn't fade within about 20 minutes, stop using the hand until it has been checked.
- **Wash** the cover, liner and shell with soap and water. Let them drain and dry; a drain hole at the front of the hand lets water out.
- **Keep it out of hot cars and direct sun** for long periods. PETG holds up far better than PLA, but heat still softens plastic.
- **Replace worn parts** by reprinting them: each finger segment, the thumb and the cover are separate files. Replace a flexor line if it frays.